

OFFUTT AFB, NE, USA

WMO: 725540

Lat: 41.117N

Lon: 95.900W

Elev: 1053

StdP: 14.15

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.4	1.5	-9.3	3.5	-0.3	-5.1	4.4	3.7	31.0	19.1	26.7	23.8	7.5	320	0.539

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	95.0	76.5	91.6	75.9	89.3	74.6	80.1	89.6	78.3	87.6	76.8	85.6	10.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.4	148.3	85.6	75.5	139.0	83.9	73.9	131.5	82.1	44.6	89.7	42.7	87.6	41.0	85.8	93.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.8	20.9	18.7	-10.7	99.7	4.4	3.4	-13.8	102.1	-16.4	104.1	-18.9	106.0	-22.1	108.4		
			-11.0	83.5	4.5	2.6	-14.2	85.4	-16.8	86.9	-19.3	88.3	-22.5	90.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.0	23.7	28.6	40.6	51.9	63.1	72.9	77.3	74.8	67.2	54.2	40.1	28.1
	DBStd	20.87	11.82	12.36	11.64	9.84	8.47	6.49	5.88	5.69	8.15	9.31	9.97	11.03
	HDD50	2920	816	603	335	92	5	0	0	0	1	63	323	682
	HDD65	5934	1281	1018	759	405	140	12	1	3	71	351	748	1145
	CDD50	3650	0	5	43	149	410	687	847	770	518	194	25	2
	CDD65	1188	0	0	1	12	81	249	383	308	137	17	0	0
	CDH74	12009	0	4	31	187	839	2517	4086	2858	1300	180	7	0
	CDH80	4739	0	2	3	51	278	980	1803	1130	449	42	1	0
Wind	WSAvg	8.5	8.8	9.0	9.8	10.3	9.0	8.1	7.0	6.9	7.6	8.4	8.7	8.5
Precipitation	PrecAvg	33.3	0.8	1.1	1.9	3.4	5.0	5.3	4.1	4.7	2.8	2.1	1.4	1.0
	PrecMax	48.4	1.7	2.7	4.8	7.4	9.6	10.9	12.0	11.5	6.8	6.2	3.1	5.0
	PrecMin	22.6	0.2	0.2	0.1	0.6	1.6	1.8	0.2	1.5	0.5	0.0	0.0	0.0
	PrecStd	6.3	0.4	0.6	1.2	1.4	2.2	2.6	2.7	2.8	1.6	1.5	0.9	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	67.4	78.6	86.3	93.1	95.4	99.9	97.8	93.0	85.6	73.2	59.8
		MCWB	44.9	53.1	59.9	64.2	70.3	75.0	77.0	76.1	73.0	67.1	58.4	51.3
	2%	DB	49.9	58.6	71.8	79.3	86.5	91.7	95.4	93.3	89.5	79.0	66.4	53.8
		MCWB	42.5	48.6	56.4	61.5	69.1	74.9	78.2	76.3	72.7	63.7	54.7	46.3
	5%	DB	44.7	51.7	65.4	74.0	82.3	89.2	92.5	89.8	85.5	73.9	61.1	48.1
		MCWB	39.1	44.0	54.1	58.4	66.9	73.6	77.6	76.1	71.6	60.8	51.8	42.5
	10%	DB	39.5	45.7	59.5	69.4	78.7	86.0	89.4	86.2	81.6	69.9	56.0	43.3
		MCWB	35.8	40.4	49.6	56.0	65.1	72.0	76.2	74.0	68.9	59.5	48.5	38.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.9	57.0	64.2	67.7	74.8	80.3	83.1	81.1	77.4	71.2	61.5	55.0
		MCDB	54.5	61.8	72.3	79.0	84.4	89.6	93.1	89.6	86.7	78.6	68.4	58.4
	2%	WB	42.9	49.2	59.8	64.0	72.2	77.6	80.9	79.3	75.2	67.4	57.3	47.3
		MCDB	49.0	57.6	67.4	74.2	82.0	87.3	90.4	88.9	84.6	74.1	64.5	51.6
	5%	WB	39.5	44.3	54.9	60.9	69.7	75.8	79.2	77.5	73.4	64.3	53.6	42.4
		MCDB	43.8	50.7	64.1	71.0	79.3	85.4	88.8	86.2	82.2	70.4	59.8	47.6
	10%	WB	36.3	40.6	50.4	57.5	67.2	74.0	77.4	75.8	71.5	60.9	49.1	39.0
		MCDB	39.6	45.7	59.4	67.2	75.9	83.2	86.3	83.7	79.3	67.6	55.7	43.1
Mean Daily Temperature Range	5% DB	MDBR	18.3	19.0	21.7	22.1	20.8	20.0	19.5	19.5	22.1	21.5	20.1	17.5
		MCDBR	26.5	27.6	32.8	31.0	26.4	23.2	23.0	23.3	24.8	27.7	27.9	25.7
		MCWBR	19.3	18.0	19.0	16.6	12.3	10.5	9.7	10.0	11.4	14.5	17.1	18.4
	5% WB	MCDBR	24.3	25.9	27.9	25.8	22.4	20.2	19.7	19.6	20.9	21.9	24.8	23.3
MCWBR		18.4	17.7	17.3	16.2	11.8	11.0	10.4	9.8	11.2	13.8	17.2	17.7	
Clear-Sky Solar Irradiance	taub	0.281	0.317	0.330	0.385	0.411	0.415	0.426	0.417	0.376	0.333	0.299	0.285	
	taud	2.459	2.372	2.400	2.263	2.262	2.303	2.266	2.303	2.371	2.478	2.501	2.457	
	Ebn at Noon	281	283	292	281	275	273	269	268	272	274	271	269	
	Edh at Noon	24	31	34	41	43	41	42	39	34	28	23	22	
All-Sky Solar Radiation	RadAvg	643	939	1217	1551	1841	2047	2059	1784	1473	1032	718	541	
	RadStd	56	57	110	107	142	158	136	110	96	120	51	54	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page